Dipendra Sharma

Software Development Engineer Intern | Algorithms & Backend

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Summary

Computer Science student with strong foundations in **Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, and OOP**. LeetCode **peak rating 1570**, **313+ problems solved**, **500-Day Badge**. Confident in Arrays, Strings, Hash Tables, Two Pointers, Sliding Window, Binary Search, Linked Lists, Stacks, Queues, and Greedy. I believe a problem well-stated is half-solved — I focus on understanding depth before writing code, and always aim for the most optimal time/space complexity first.

Competitive Programming & Problem Solving

LeetCode: Peak Rating **1570** | **313+ problems solved** | **500-Day Badge** | 3+ contests
Patterns Mastered: Kadane's, Two-Pointer, Sliding Window, Binary Search on Answer, Dutch National Flag, Prefix Sum, Hash Map Counting, Monotonic Stack, Fast/Slow Pointers
Approach: Analyze time/space complexity first | target O(N) or O(log N) after brute force | iterate to optimal

Technical Skills

Languages: C, C++, Java, JavaScript, TypeScript, SQL
Core CS: **OS** (Processes, Threads, Scheduling, Memory), **DBMS** (Normalization, ACID, Indexing, Transactions), **CN** (TCP/UDP, HTTP, DNS), **OOP** (Polymorphism, Inheritance, Encapsulation, Abstraction)
Linux: Kernel fundamentals, systemd, cron, SSH, bash scripting, log analysis, networking tools (ss, netstat)
Backend: Node.js, Express.js, REST APIs, JWT Authentication, RBAC, PostgreSQL, Prisma, MongoDB
Frontend: React, Next.js, TypeScript, Redux, Tailwind CSS
DevOps: Docker, Git, GitHub Actions, AWS (EC2, S3, IAM), Vercel, Render

Projects

Daigo — Peer-to-Peer Commute Marketplace | GitHub **Aug 2026 – Present**

- Engineered **multi-hop relay matching** using graph pathfinding, chaining two drivers through a single transfer point to optimize route coverage and minimize detours.
- Implemented geospatial route matching using **raw PostGIS queries** (via Prisma `$queryRaw`) and indexed spatial data to enforce detour limits and calculate suggested meeting points.
- Built concurrency-safe real-time accept flows using **Prisma transactions** to prevent race conditions in first-accept-wins driver matching; implemented Redis pub/sub for distributed WebSocket delivery.

Tech: Next.js, TypeScript, PostgreSQL + PostGIS, Prisma, Redis, Stadia Maps

Yeti Jobs | GitHub | Live **Jan 2026 – Apr 2026**

- **Built 50+ REST APIs** with RBAC — hash-map based role validation for O(1) permission checks across admin, recruiter, and candidate workflows.
- Optimized PostgreSQL via composite indexing (tree-structured search), cutting query time **30%** (7ms → 5ms per EXPLAIN ANALYZE); connection pooling handled **20 concurrent DB connections**.
- Load-tested with Apache Bench: **1,000 requests** at **100 concurrent users → 67 req/sec**, p95 **1,929ms**, **0 failures** on a free-tier Render instance.
- Multi-stage Docker builds reduced image size by **76%**; cron job eliminated cold starts (**50s → <2s**), sustaining **99.5% uptime** over 3 months. Deployed via GitHub Actions CI/CD.

Tech: PostgreSQL, Node.js, Express, React, TypeScript, Docker, JWT, Jest, GitHub Actions

Achievements

- **CodeaThon 7.0 Winner** — solved 4/5 algorithm questions, 1st place (cash prize, 120+ participants).
- LeetCode peak rating **1570** — 313 problems solved, 500-Day Badge, 100-Day Badge 2026.
- Selected Contributor — **GirlScript Summer of Code** 2026 (open-source).
- Led multiple hackathon teams of 3-6 (SIH, Digimantra, Nepal Police) — shipped prototypes under 24-hour deadlines.
- **GitHub**: 2000+ commits (last 12 months) | **5+ merged PRs** including **Zed editor** contributions.
- **CEED Member** — Entrepreneurship Program, Chitkara University.

Certifications

- **AWS Educate Introduction to Cloud 101** — EC2, S3, IAM fundamentals
- **C++ Code Like You Are in MATRIX** — Memory Management, Data Structures, Algorithms

Education

Chitkara University **Aug 2024 – Expected 2027**

Bachelor of Computer Applications (BCA) ***CGPA: 9.38/10***

- Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Software Engineering